

Experiment No. 16

Title

Feed Forward Neural Network Using Artificial Intelligence

Aim

To develop a Python program to implement a Feed Forward Neural Network for classification.

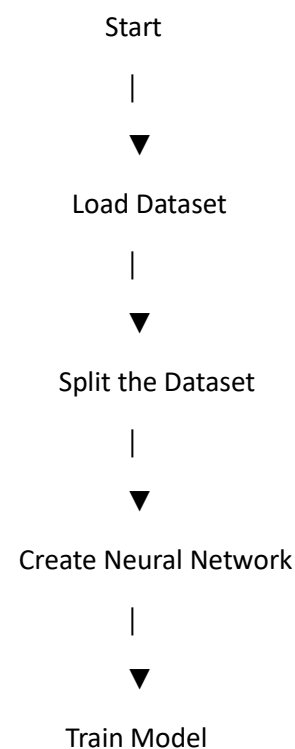
Objective

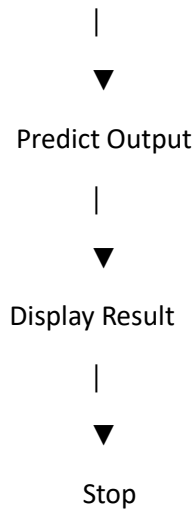
The objective of this experiment is to understand the working of a Feed Forward Neural Network and use it to classify data using input, hidden, and output layers.

Algorithm

- Step 1:** Import the required libraries.
 - Step 2:** Load the dataset.
 - Step 3:** Split the dataset into training and testing sets.
 - Step 4:** Create the Feed Forward Neural Network model.
 - Step 5:** Train the model using the training data.
 - Step 6:** Predict the output for the test data.
 - Step 7:** Display the predicted results.
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Flowchart





Python Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Feed Forward Neural Network
model = MLPClassifier(hidden_layer_sizes=(10,),
                      max_iter=1000,
                      random_state=42)

# Train model
model.fit(X_train, y_train)
```

```
# Predict
prediction = model.predict(X_test)

print("Predicted Values:")
print(prediction)
```

Sample Output

Predicted Values:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
```

Result

The Feed Forward Neural Network successfully classified the given dataset.

Conclusion

Thus, the Feed Forward Neural Network was successfully implemented for classification. The model learned from the training data and accurately predicted the class labels for the test dataset.